Ex. No: 8 Nmap to Discover live hosts. using Nmap
Date: 26/8/26 Scans on the Tryhackme platform.

AIM:
To discover live hosts using Nmap scans (ARP, ICMP, TCP/UDP) on the Tryhackme platform room link, https, Tryhackme.com/room/nmapcr

procedure:

1. Task:
ARP scan: this scan uses ARP requests to discover live hosts.
ICMP scan: this scan uses ICMP request to identify live hosts.
TCP/UDPping scan: this scan sends packets to TCP ports and TCP ports UDP ports to determine live hosts.

2. Subnetwork:
network segments It is a collection of computers linked together trough a common medium.
Subnetwork It is usually the equivalent of one. or more networks segments linked together and. configured. to utilized the same routes in an IP network.

3. Enumerbeiy targets:
We need to identify the targets we want to scan before we conduct any "action"

A. Nmap Host Discovery using ARP.
1) When a privilaged user has to scan LAN, NMAP uses ARP Requests.
2) to discover ethernet hosts without port scanning- live systems use: nmap -sn targets.

3) nmap -PR -sn <Target>

-PR specifies that you only want an ARP.

4) nmap -PR -sn <machine-IP> /24

5) Nmap host discovery using ICMP.

1) Add -PE option to utilize ICMP echo requests to discover live hosts.

2) Nmap -P sn Target to check host it live.

3) -PP option inmuts nmap to use icmp timestamp requests.

4) -pm option can enable this scan to ICMP address mark requests.

nmap -PM -sn <:targets>

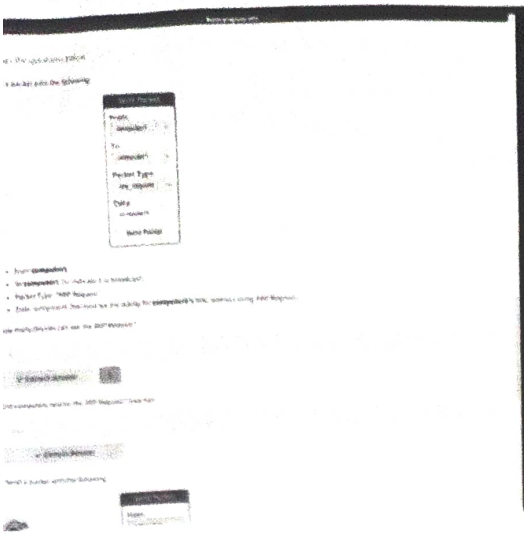
6) Nmap host discovery using TCP and UDP

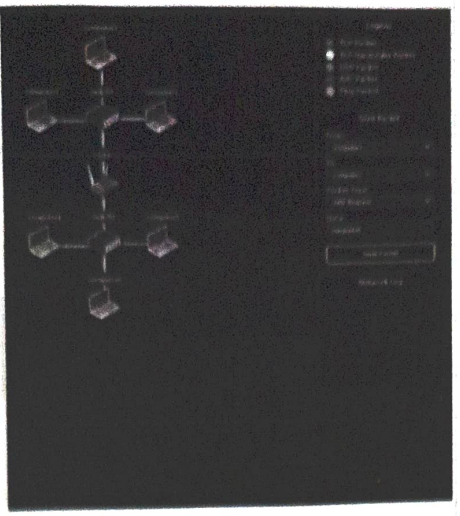
1) Nmap to use TCP SYN ping use -PS followed by port number, range, list (or) combination of this -PS21, -PS25, taughts ports 21, 22, 23, 24, 25.

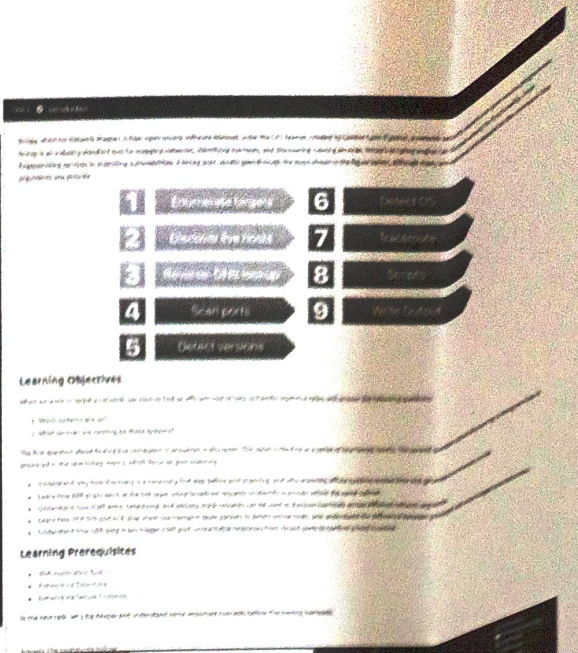
2) nmap -PS -sn TARGET check the TCP port is of

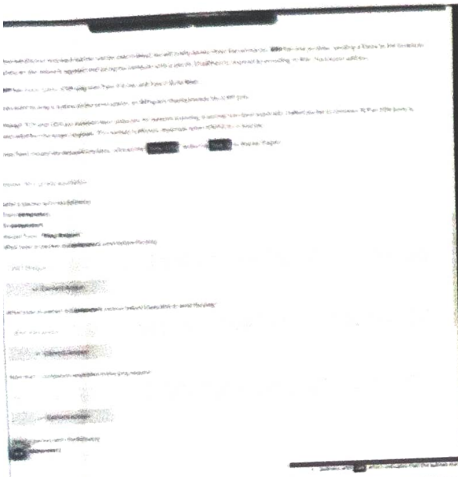
3) TCP ACK ping you must be running nmap as a privilged users.

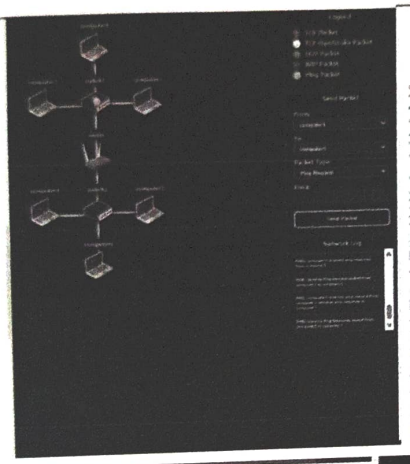
4) nmap -PA -sn Target to discuss online hosts on the targets subnet.

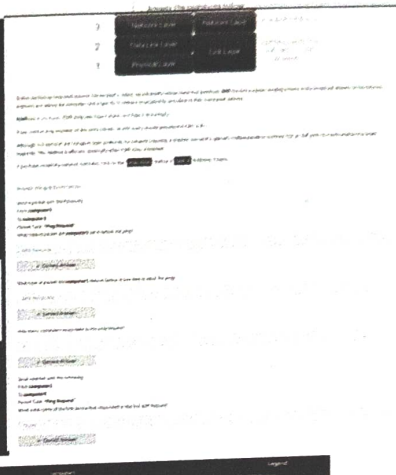


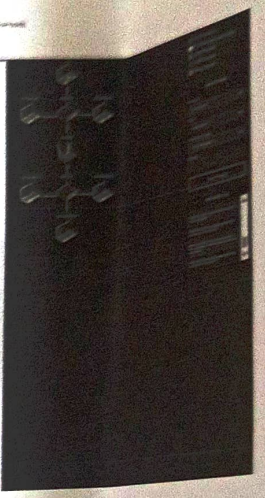


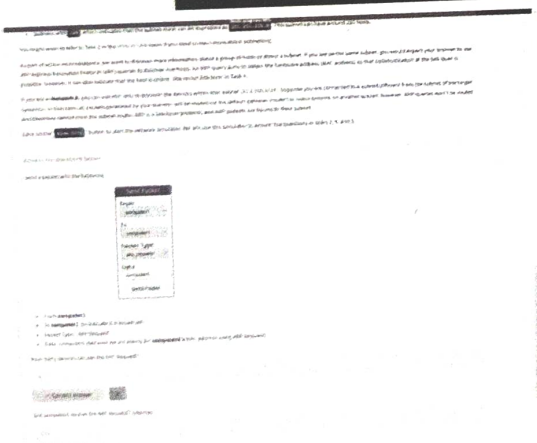

Learning Objectives
Learning Prerequisites

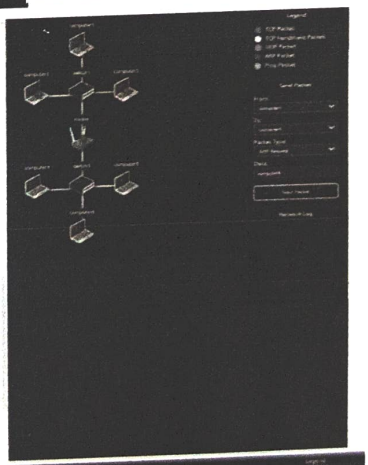


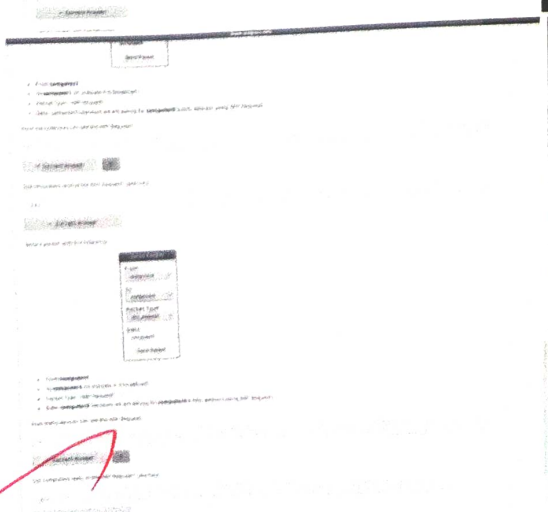


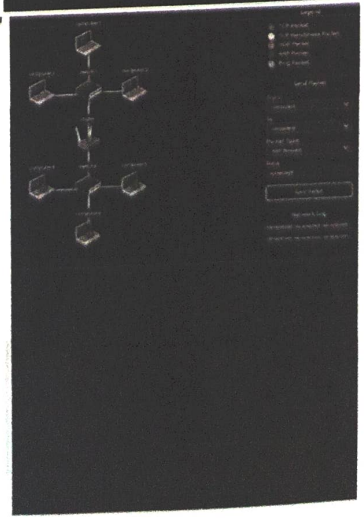












Room progress: 63%

Start | Extend (52min left) | Download VPN Profile

Task 1 Introduction

Task 2 Subnetworks

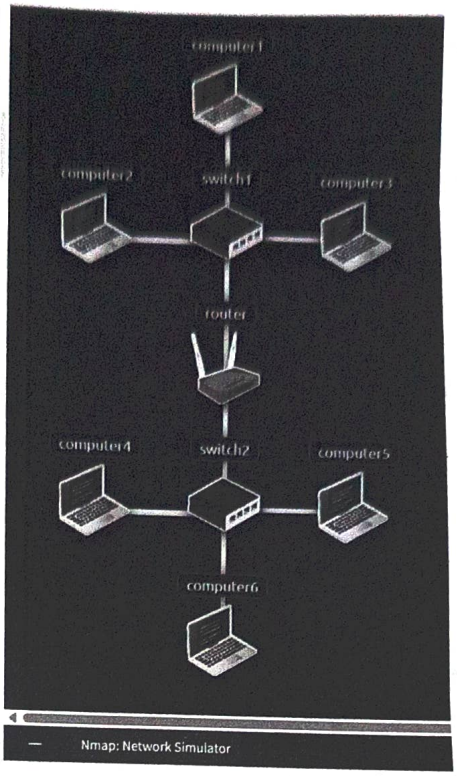
Task 3 Understanding Hosts Discovery through TCP/IP Layer

Task 4 Enumerating Targets

Task 5 Nmap Host Discovery Using ARP

Task 6 Nmap Host Discovery Using ICMP

We can ping every IP address on a target network and see who would respond to our `ping` (ICMP Type 8/Echo) requests with a ping reply (ICMP Type 0). Simple, isn't it? Although this would be the most straightforward approach, it is not always reliable. Many firewalls block ICMP echo; new versions of MS Windows are configured with a



Nmap: Network Simulator

Result:

Thus, the Nmap to discover live hosts, using Nmap Scans on the TryHackMe platform.